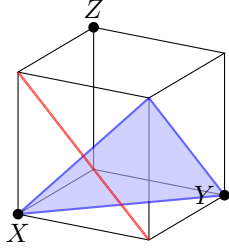


1 Exercise 1

consider the cube $C = (x, y, z) \in \mathbb{R}^3 | 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$ in $\mathbb{R}^3$. Find the angle between the plane P_1 containing the 3 points: $(0, 1, 0), (1, 0, 0), (1, 1, 1)$ and the plane P_2 containing the 3 points: $(0, 0, 0), (1, 1, 0), (1, 0, 1)$.



To find the angle between the two planes, we must firstly find the normal vector that the individual planes define and then find the angle between those two.

$$P_{1X} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad P_{1Y} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad P_{1Z} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Vectors for P_1

$$P_{1XY} = P_{1X} - P_{1Y} = \begin{pmatrix} 0-1 \\ 1-0 \\ 0-0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad P_{1XZ} = P_{1X} - P_{1Z} = \begin{pmatrix} 0-1 \\ 1-1 \\ 0-1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

The normal vector for plane P_1 is the cross product of the two vectors in the plane.

$$\vec{N1} = P_{1XY} \times P_{1XZ} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

The procedure goes for the second set of points, to generate the set of vectors, the plane they sit on, and the normal vector that extends from it.

Points

$$P_{2X} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad P_{2Y} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad P_{2Z} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

Vectors for P_2

$$P_{2XY} = P_{2X} - P_{2Y} = \begin{pmatrix} 0-1 \\ 0-1 \\ 0-0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}, \quad P_{2XZ} = P_{2X} - P_{2Z} = \begin{pmatrix} 0-1 \\ 0-0 \\ 0-1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

Normal vector from plane P_2

$$\vec{N2} = P_{2XY} \times P_{2XZ} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

The angle between these two vectors will then indicate the angle between the planes P_1 and P_2

$$\theta = \cos^{-1}\left(\frac{N1 \cdot N2}{|N1||N2|}\right) \Rightarrow \frac{(-1)(1) + (-1)(-1) + (1)(-1)}{\sqrt{(-1)^2(-1)^2(1)^2} \cdot \sqrt{(1)^2(-1)^2(-1)^2}} \Rightarrow \frac{-1}{\sqrt{(3)^2}} \Rightarrow \cos^{-1}\left(\frac{-1}{3}\right) \approx \frac{180}{\pi} \approx 109.492$$

2 exercise 2

Find the center of mass $(\bar{x}, \bar{y}, \bar{z})$, for the wire C shaped as a helix, with density $p = kz$, (k is a constant). The wire C has parametrization:

$$x(t) = 3\cos(t), y(t) = 3\sin(t), z(t) = 4t, 0 \leq t \leq \pi$$

The following equations define the center of mass:

- The curve length $s(t)$, as a function of t satisfies: $\frac{ds}{dt} = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$
- $m = \int_C p(x, y, z) ds$
- $\bar{x} = \frac{1}{m} \int_C xp(x, y, z) ds, \bar{y} = \frac{1}{m} \int_C yp(x, y, z) ds, \bar{z} = \frac{1}{m} \int_C zp(x, y, z) ds$

This is a rather convoluted question, essentially asking, if we have a piece of wire, in this case a helix, meaning there is a one large curve, since it is bounded to $0 \leq t \leq \pi$, what is the position in space of the center of its mass. The parametrization describes how the *coil* will expand if we increase t . The first two components x and y are trigonometric functions, causing the direction to be circular around the z -axis, and the z component simply indicates that it increases linearly with t .

Firstly we need to calculate the change in t into changes on the actual physical wire. We're given the density p and mass is the product of density and length. So the curve length is calculated as:

$$\frac{ds}{dt} = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$$

differentiating the functions

$$x'(t) = \frac{d}{dt}x(t) \Rightarrow -3\sin(t)$$

$$y'(t) = \frac{d}{dt}y(t) = \frac{d}{dt}3\cos(t)$$

$$z'(t) = \frac{d}{dt}z(t) = \frac{d}{dt}4t$$

into the curve length

$$\frac{ds}{dt} = \sqrt{(-3 \cdot \sin(t))^2 + (3 \cdot \cos(t))^2 + (4)^2}$$

rearrange, use distributive property and recognize the idiot formula $\sin^2(t) + \cos^2(t) = 1$

$$= \sqrt{9 + 1 + 16} = \sqrt{25} = 5$$

$$ds = 5dt$$

Relationship for density is given, with z given and k is some constant, it can be rewritten as:

$$p = k \cdot z$$

$$p = 4tk$$

Substituted into the mass relationship, with a change of variable from the $\frac{ds}{dt}$ relationship, so that mass is now in terms of t :

$$m = \int_0^\pi (4tk)(5)dt \Rightarrow 10k\pi^2$$

Since the lower bound is zero and are a single term with factors, it gets reduced to zero. Each of the coordinates for the center of mass can now be evaluated.

$$\begin{aligned} \bar{x} &= \frac{1}{10k\pi^2} \int_0^\pi 3 \cdot \cos(t) \cdot 4t \cdot k \cdot 5dt \\ &= \frac{60k}{10k\pi^2} \int_0^\pi t \cdot \cos(t)dt \Rightarrow \frac{60k}{10k\pi^2} [t \cdot \sin(t) + \cos(t)]_0^\pi \end{aligned}$$

$$= \frac{6}{\pi^2} \cdot (-1) - \frac{6}{\pi^2} + \cos(0) = -\frac{12}{\pi^2}$$

The same can be done for the two other coordinates:

$$\begin{aligned}\bar{y} &= \frac{1}{10k\pi^2} \int_0^\pi 3 \cdot \sin(t) \cdot 4t \cdot k \cdot 5dt \\ &= \frac{60k}{10k\pi^2} \int_0^\pi t \cdot \sin(t) dt \Rightarrow \frac{6}{\pi^2} [\sin(t) - t \cdot \cos(t)]_0^\pi \Rightarrow \frac{6}{\pi} \\ \bar{z} &= \frac{1}{10k\pi^2} \int_0^\pi (4t) \cdot 4t \cdot k \cdot 5dt \\ &= \frac{80k}{10k\pi^2} \int_0^\pi t^2 dt \Rightarrow \frac{8}{\pi^2} [\frac{t^3}{3}]_0^\pi \Rightarrow \frac{8\pi}{3}\end{aligned}$$

Giving coordinates:

$$[\bar{x}, \bar{y}, \bar{z}] = [-\frac{12}{\pi^2}, \frac{6}{\pi}, \frac{8\pi}{3}]$$

3 exercise 3

Let F be a vector field on $\mathbb{R}^3$ given by:

$$F(x, y, z) = (0, 0, 1)$$

Using surface integrals, show that the flux through the upper hemisphere of the unit sphere:

$$S_+^2 = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 + z^2 = 1, z \geq 0\}$$

is the same as the flux through the unit disk in the xy -plane

$$D = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 \leq 1, z = 0\}$$

we're given a vector field that essentially shows an upwards flow in in the Z direction, with a magnitude of 1.

$$F(x, y, z) = (0, 0, 1)$$

As for the hemisphere we're bounded for $z \geq 0$, therefore we can solve the hemisphere equation for z , essentially telling us the height z at any point in the sphere over (x, y) on the disc underneath.

$$x^2 + y^2 + z^2 = 1 \Rightarrow z = \sqrt{1 - x^2 - y^2}$$

Then we can establish what the flux is at any tiny piece in the hemisphere. This can be expressed as a tiny patches on the hemisphere. If we imagine we're looking down along the Z -axis onto the XY -plane, each of the small patch on the hemisphere, casts a project or shadow onto the XY -disk. These two surfaces must be of equal size. This means that:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS = \iint_D dA$$

In other words, the entire hemisphere projects onto the disk. The unit disk is given as:

$$D = x^2 + y^2 = 1$$

And because it is circular, polar coordinates can be used to express it. And with a unit disk, the bounds are $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$. The changes in coordinates $x = r\cos(\theta)$, $y = r\sin(\theta)$, results in every little area becoming $dA = r dr d\theta$. The so the expression for the flux related to the disk becomes:

$$\Phi = \int_0^{2\pi} \int_0^1 r dr d\theta$$

The extra r , introduced here, is to compensate for the arc length. Evaluating Φ

$$\begin{aligned} \Phi &= \int_0^{2\pi} \int_0^1 r dr d\theta \Rightarrow \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^1 d\theta \\ &\Rightarrow \int_0^{2\pi} \frac{1}{2} d\theta \Rightarrow \left[\frac{\theta}{2} \right]_0^{2\pi} = \frac{2\pi}{2} = \pi \end{aligned}$$

To calculate the flux through the disk.

Using the same field $F = (0, 0, 1)$, we need the normal vector to the surface of the disk in the same orientation as the vector field meaning $\mathbf{n} = (\hat{0}, 0, 1)$. The dot-product of these two vectors becomes $\mathbf{F} \cdot \hat{\mathbf{n}} = 1$, meaning they point in the same direction. This lets us create the integral of the disk flux.

$$\iint_D \mathbf{F} \cdot \hat{\mathbf{n}} dA = \iint_D 1 dA = \int_0^{2\pi} \int_0^1 r dr d\theta = \pi$$

This can also be shown when the hemisphere gets parameterized as $r(x, y) = (x, y, f(x, y))$ where $f(x, y) = \sqrt{1 - x^2 - y^2}$. We can then create two small vectors, representing small movement along the surface of the hemisphere $r_x = (1, 0, f_x), r_y = (0, 1, f_y)$. The cross product of those two, will then be perpendicular to them, as it'll become the normal vector to that tiny surface patch. The dot-product of this and the vector field F yields:

$$\mathbf{F} \cdot (r_x \times r_y) = 1$$

meaning that, that the vector picks out the z-component of the vector field. This leads to

$$\Phi_S = \iint_D 1 dA = \pi$$

4 exercise 4

let ϕ be a scalar field and F a vector field given by:

$$\phi(x, y, z) = xyz^2$$

and

$$F(x, y, z) = (x^2, y, z)$$

- Find $\nabla\phi$ and $\nabla \cdot F$ and $\nabla \cdot (\phi F)$
- try to deduce a general expression for $\nabla \cdot (\phi F)$ expressed by $\nabla\phi$ and $\nabla \cdot F$

Given the scalar field, the gradient ∇ can be found by differentiating once in each direction. The gradient of a scalar field is a vector field.

$$\nabla\phi = \begin{bmatrix} \frac{\partial\phi}{\partial x} \\ \frac{\partial\phi}{\partial y} \\ \frac{\partial\phi}{\partial z} \end{bmatrix} = \begin{bmatrix} yz^2 \\ xz^2 \\ 2xyz \end{bmatrix}$$

The next is finding the divergence of a vector field, which turns it into a scalar field.

$$\nabla F = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = 2x + 2$$